12. Information

We are approaching the conclusion of our journey. In the previous few chapters, I spoke about the concrete applications of quantum gravity: the description of what happened to the universe around the time of the Big Bang; the description of the properties of the heat of black holes and the suppression of infinity.

Before concluding, I would like to return to the theory, but looking at its future, and to speak about information: a spectre that is haunting theoretical physics, arousing enthusiasm and confusion.

This chapter is different from the preceding ones, where I spoke of ideas and theories not yet tested but well defined; here, I'm speaking of ideas still confused, badly in need of organization. If, dear reader, you have found the journey so far a little rough, then hold on tighter, because we're now flying between voids of air. If this chapter seems particularly opaque, it's not because your ideas are confused. It's because the one with the confused ideas is me.

Many scientists suspect today that the concept of 'information' may turn out to be a key for new advances in physics. Information is mentioned in the foundations of thermodynamics, the science of heat, the foundation of quantum mechanics and in other areas besides, with the word quite often used very imprecisely. I believe there is something important in this idea. I'll try to explain why, and to show what information has to do with quantum gravity.

Before anything else, what *is* information? The word 'information' is used in common parlance to mean a variety of different things, and this imprecision is a source of confusion in science as well. The scientific notion of information, however, was defined with clarity in 1948, by the American mathematician and engineer Claude Shannon, and is something very simple: information is the measure of the number of possible alternatives for something. For example, if I throw a die, it can land on one of six faces. When we've seen it fall on a particular one of these, we have an amount of information $N = 6$, because the possible alternatives are six in number. If I don't know which day of the year is your birthday; there are 365 distinct possibilities. If you tell me the date, I have the information N = 365. And so on.